

4

Reductionism

In the present chapter, I defend two main theses, as against the views of ethical reductionists: that at least some evaluative properties are irreducible, and that moral knowledge cannot be explained without appeal to ethical intuition.

4.1 What is reductionism?

Both ethical naturalism and subjectivism are forms of *reductionism* about ethics. Reductionism holds that all evaluative properties are reducible to natural properties.

What does that mean? I take ‘natural properties’ to include all properties (including relational properties) that can be referred to using non-evaluative terms.¹ This includes both physical properties (such as mass, shape, and electric charge) and mental properties (such as believing, feeling sad, and having a tickling sensation). Naturalists and subjectivists disagree with each other over whether certain sorts of mental properties—namely, observers’ attitudes towards objects of evaluation—should be included among those to which value properties reduce; but they agree that value properties reduce to *some* sort of natural properties. To say that an evaluative property ‘reduces’ to some collection of natural properties is to say that one could explain the nature of the evaluative property—explain *what it is*—entirely in terms of those natural properties.

Initially, reductionism might seem paradoxical: how could wholly *non*-evaluative terms be used to explain the nature of value? To see that this is not so paradoxical, consider an analogy: some say that for an object to be red is just for it to reflect predominantly light with

wavelengths in the neighborhood of 650 nanometers. Notice that this explanation is given entirely using non-color terms ('reflect', 'wavelength', etc.), so that a person with no previous awareness of color could, if the explanation is correct, come to understand the nature of redness on the basis of it. Whether or not this account of the nature of redness is correct, it is at least not *obviously* false. Similarly, perhaps it is possible to give an explanation of what goodness is which will not use any evaluative terms.²

A key advantage often claimed for reductionism is that it holds the promise of rendering our *knowledge* of morality intelligible. If moral properties are irreducible, non-natural properties, then it seems that the only way we might be aware of them is by 'intuition'—a faculty that contemporary philosophers have been reluctant to embrace. But the view that moral properties are reducible to natural properties holds out the hope that we might know moral facts in a manner similar to how we know (other) natural facts, such as by observation or scientific reasoning.³

In the remainder of this chapter, I argue both that reductionism about ethics is false, and that this alleged advantage of the theory is illusory: reductionists cannot escape the appeal to intuition as a source of moral knowledge.⁴

4.2 Analytic reductionism

Analytic reductionism holds that the meaning of any moral term can be given using non-moral terms. For example, an analytic reductionist might hold that 'x is good' is synonymous with 'x increases the total amount of enjoyment in the world'. Or one might hold that 'x is good' is synonymous with 'x is something that we desire to desire'.⁵

G. E. Moore's Open Question Argument refutes analytic reductionism.⁶ Suppose '*N*' is an expression containing only non-evaluative terms that denotes some property. 'Are things that have *N* good?' is an *open question*. What this means is that a person could coherently give either answer to the question. But 'Are good things good?' is not an open question; a person could not coherently give either answer to *that* question. Therefore, the two questions are not equivalent. Therefore, 'good things' is not synonymous with 'things that have *N*'. To illustrate, consider the theory that 'x is good' means 'x increases the amount of enjoyment in the world'. Moore would argue:

1. 'Is it good to do what increases the amount of enjoyment in the world?' is an open question.
2. 'Is it good to do what is good?' is not an open question.

3. Therefore, the questions in (1) and (2) are not synonymous.
4. Therefore, 'increases the amount of enjoyment in the world' is not synonymous with 'is good'.

The idea behind premise (1) is that a person could coherently wonder whether increasing enjoyment is good (after all, an ascetic value system is logically consistent). But as (2) observes, one could not wonder whether doing good is good; no coherent value system denies that doing good is good.

The idea behind (3) is that since one question is open and the other is not, they are not the same question. One might also simply see immediately that the two questions are different, even without invoking the notion of 'open questions'.

Step (4) follows from (3) because the questions in (1) and (2) differ only in that one of them substitutes the predicate 'is good' for the predicate 'increases the amount of enjoyment in the world'; therefore, the difference in meaning between the two questions must be traced to a difference in meaning between those two predicates. If the two predicates had the same meaning, as the reductionist claims, one should be able to substitute one for the other in any sentence without changing the meaning of the sentence.

Moore says that a similar argument will work against *any* analytic reductionist theory. I know of one apparent exception to this: consider an analytic ideal observer theory, which holds that 'x is good' means something like, 'An ideally rational, knowledgeable, sensitive being would approve of x'. The Open Question Argument may fail against this theory, because the first premise is not clearly true.⁷ That is, it is not clear that questions like 'Are the things that an ideally rational, informed, sensitive being would approve of really good?' are open questions. One may think that, provided we characterize ideal observers correctly, an ideal observer is bound to get things right.

I think that there are two ways of understanding this sort of theory, one that makes it non-reductionist and immune to the Open Question Argument, and another that makes it reductionist but vulnerable to the Open Question Argument. On the one hand, if 'ideal rationality' is taken to include the capacity for making rational evaluations, 'ideal knowledge' is taken to include moral knowledge, or 'ideal sensitivity' is taken to include sensitivity to the moral features of things, then I think 'Are the things that an ideally rational, informed, sensitive being would approve of really good?' is not an open question. But in that case, I think we have a non-reductionist theory, since the existence of other evaluative facts or

features must be presupposed in defining 'good'. I am not concerned here to refute non-reductionist accounts of value.

If, on the other hand, ideal rationality only includes rationality in the making of non-moral judgments, ideal knowledge only includes non-moral knowledge, and ideal sensitivity only includes non-moral sensitivity, then I think it is an open question whether an ideally rational, knowledgeable, and sensitive being would also possess correct moral beliefs; therefore, it is an open question whether what such a being approved of would be good. Conceivably, a being might be perfectly rational, informed, and sensitive in all non-evaluative areas and yet lack any correct moral judgments. In such a case, this being might fail to approve of what is good, or even approve of things that are bad.

I believe, then, that Moore's argument succeeds against all genuinely reductionist theories of the meanings of evaluative expressions. Reductionists typically grant that *if* the Open Question Argument succeeds against such theories as the theory that 'good' means 'conducive to pleasure', then it succeeds against all analytic reductionist theories. However, they have said that Moore's argument fails even against the theory that 'good' means 'conducive to pleasure', for at least two reasons.

First, some say that Moore begs the question. Since the reductionist we're confronting holds that the good is just whatever increases enjoyment, obviously he will say that it is *not* an open question whether it is good to do what increases enjoyment. For Moore to assert otherwise just assumes that the reductionist theory is false.⁸

This response employs an overly expansive conception of 'begging the question'. It is not the case that whenever an argument deploys a premise that directly and obviously contradicts an opponent's position, the argument begs the question. Still less is it true that whenever a consistent opponent would reject at least one of an argument's premises, the argument begs the question. (The latter condition applies to every valid argument.) Consider another famous philosophical argument: Philosopher *A* claims that 'knowledge' means 'justified, true belief'. Philosopher *G* points out the following sort of example:

Suppose that Smith justifiably believes that Jones owns a Ford (he has often seen Jones driving a Ford, has seen the title to the car, and so forth). Smith correctly infers from this that the following statement is also true: 'Jones owns a Ford, or Brown is in Barcelona'. (Barcelona was selected randomly; Smith has no idea where Brown is.) But suppose that, improbably enough, Jones actually

does *not* own a Ford; it was sold just a few minutes ago. But by pure coincidence, Brown happens to be in Barcelona. In this case, Smith believes [Jones owns a Ford, or Brown is in Barcelona]; he is justified in believing this (since he correctly inferred it from a justified belief); and it is true (since Brown is in Barcelona). But intuitively, Smith does not *know* [Jones owns a Ford or Brown is in Barcelona]. Thus, justified, true belief is not the same as knowledge.

The above argument is widely, and rightly, taken to conclusively refute the 'justified, true belief' account of knowledge; indeed, it is one of the few, and one of the most celebrated, examples of a conclusive refutation of a previously widely-held view in modern philosophy.⁹ How weak it would be for *A* to reply:

G has merely begged the question. I say that knowledge is justified, true belief. From this, it directly and obviously follows that Smith *does* know [Jones owns a Ford or Brown is in Barcelona] in *G*'s example. For *G* to assert that Smith lacks such knowledge just assumes that my definition is wrong. So *G* has proven nothing.

A is correct to note that *G*'s premise directly and obviously contradicts *A*'s theory. But this does not mean that *G* begged the question; it means only that *G*'s refutation of *A* was direct and obvious. *G* succeeds while *A* fails because *G*'s premise, once stated, is intuitively obvious, while *A*'s theory is not intuitively obvious but needs to be tested by considering examples. *A* thus is not in a position to simply appeal to his theory as a justification for denying the premise that would be used to refute the theory.

Similarly, Moore succeeds against the reductionist because Moore is able to adduce premises that, once stated, are intuitively obvious to competent English speakers, whereas the reductionist's proposed analysis of 'good' is not intuitively obvious. We have a sense of when a statement is contradictory, and 'Enjoyment isn't good' does not sound contradictory. The reductionist thus is not in a position to simply appeal to his theory to justify rejecting the premises Moore would use against it. Nor should the reductionist question the general reliability of our judgments of consistency and inconsistency. For judgments of this kind are the essence of our knowledge of logic, without which none of us are in a position to advance or evaluate any arguments.

The second reply to Moore begins with the observation that even

for individuals who understand a given word, it is often difficult off-hand to identify the correct definition of that word. For instance, one who knows the word 'circle' may still have difficulty coming up with the definition, 'the set of all points a given distance from a given point'. Thus, a question may often appear to someone to be open even when it is not, simply because the person does not realize what the correct analysis of his own concept is. Perhaps Moore's question, 'Is it good to do what increases the amount of enjoyment in the world?' only appears open to us because we do not at first realize what the correct analysis of 'good' is.¹⁰

This response rests on an uncharitable reading of Moore. Moore need not assume that the correct analysis of a given concept is always immediately obvious to anyone who has the concept. Instead, Moore can get by with only the assumption that, once a specific definition has been stated, and once competent speakers of the language have thoroughly understood and reflected carefully on that definition, their linguistic intuitions provide at least a *prima facie* guide to the definition's correctness. This assumption is much weaker and more reasonable than the assumption that the analysis of one's concepts is always immediately obvious, and it is difficult to see how anyone engaged in attempting to define words or concepts—as the analytic reductionist is—can reject it. For without it, it is hard to see how we could ever know what the definition of anything is.

The intuitions to which Moore appeals—for example, that 'Pleasure is not good' is consistent, and that 'Is pleasure good?' does not mean 'Is pleasure conducive to pleasure?'¹¹—are ones that occur after competent speakers of English have thoroughly understood the definition of 'good' under discussion and have reflected carefully on it. Therefore, these intuitions provide at least a *prima facie* test of the definition's correctness. That is, in the absence of specific, independent grounds for rejecting these intuitions or holding them to be unreliable, they provide adequate reason for rejecting the reductionist's definition. Since reductionists have in fact provided no such grounds, we should reject the proposed definition.¹²

Moore's claim to refute *all* forms of analytic reductionism rests on his ability to produce similar intuitions for every proposed definition of 'good'. Consider some other potential definitions, and some premises that might be used in Moorean arguments against them:

Proposed Analysis

x is good = God approves of x .

Moorean Premise

'God is good' does not mean 'God approves of himself'.

x is right = x is in accord with the prevailing social customs.	'Is it right to obey social customs?' does not mean 'Does obeying social customs accord with social customs?'
x is good = I approve of x .	'Are the things I approve of really good?' does not mean 'Do I really approve of the things I approve of?'
x is good = x promotes life. ¹³	'Is life good?' does not mean 'Does life promote itself?'

I cannot discuss every possible analysis of 'good' here. But the above examples (including the case of the definition involving enjoyment) suggest that Moore could indeed adduce relevant, plausible premises for his argument against any form of analytic reductionism. I am not making an inductive argument based on five cases here. Rather, I am suggesting that upon considering these five cases, one can see a pattern, a method Moore could use to generate Open Question-type arguments in further cases.

4.3 The is-ought gap

4.3.1 Hume's law: an initial statement

Closely related to Moore's refutation of analytic reductionism is another doctrine in metaethics known as that of the 'is-ought gap', or 'Hume's Law'. This is the idea that it is impossible to validly derive an evaluative conclusion from wholly non-evaluative premises—or, as it is sometimes said, one cannot deduce conclusions about what *ought* to be from premises about what *is*. The *locus classicus* of this idea is the following well-worn passage from David Hume:

In every system of morality, which I have hitherto met with, I have always remark'd, that the author proceeds for some time in the ordinary way of reasoning, and establishes the being of a God, or makes observations concerning human affairs; when of a sudden I am surpriz'd to find, that instead of the usual copulations of propositions, *is*, and *is not*, I meet with no proposition that is not connected with an *ought*, or an *ought not*. This change is imperceptible; but is, however, of the last consequence. For as this *ought*, or *ought not*, expresses some new relation or affirmation, 'tis necessary that it shou'd be observ'd and explain'd; and at the same time that a reason should be given, for what seems

altogether inconceivable, how this new relation can be a deduction from others, which are entirely different from it.¹⁴

Note that Hume is not merely observing that from 'x is the case' one cannot validly infer 'x ought to be the case'. Rather, he is making the general point that *no* collection of facts about what is the case entail *any* claim about what ought to be. Another way of stating the principle is to say that any (non-evaluative) description of the world can consistently be combined with any value system; or even with nihilism.¹⁵

Consider a simple challenge to Hume's Law. Suppose I pronounce that communism is bad. You ask me why I believe this. I say: 'It caused tens of millions of deaths, widespread poverty, and a great deal of human suffering. It is reasonable to believe that it would continue to have such effects in the future. Therefore, it is a bad system of government'. Assume that my historical claim is adequately established, and assume that you accept my generalization that communism can be expected to continue to have similar effects. Is not my conclusion thereby justified? Surely it is not merely some sort of simple logical fallacy I have committed. Yet the premises of my argument are entirely non-evaluative, while the conclusion is evaluative.

According to the standard doctrine, which I endorse, the argument fails to bridge the is-ought gap because it contains certain *implicit premises* which are evaluative. Specifically, the argument implicitly assumes that life is good and that poverty and human suffering are bad—if we do not assume these things, then the conclusion that communism is bad does not follow. Now, these assumptions may be perfectly reasonable, but they are, nevertheless, evaluative premises; therefore, the argument in question is not an example of a derivation of an evaluative conclusion *from wholly non-evaluative premises*.¹⁶ A similar point applies to all of the familiar ways in which we typically argue for evaluative conclusions; we generally argue that x is good by presupposing some evaluative standards. If we know that y is good, then we may argue that x is good by showing that x leads to y. If we know that people should do A when in circumstances of type C, then we may argue that S should do A by showing that S is in C.

This shows that some familiar moral arguments fail to bridge the is-ought gap. But what justifies the general conclusion that no argument can possibly do so? I think the initial justification for Hume's Law is intuitive: it simply is very difficult to see, on the face of it, how some mere description of how the world in fact is (in non-

evaluative respects) could rule out some otherwise possible value system. It is equally difficult to see how a nihilist who recognized the natural properties of things but failed to value anything could be convicted of contradicting himself; it seems that nihilism would always remain a coherent position in the face of any purely natural facts. I do not say that it would be a *plausible* or *justified* position, but it would be coherent.

Another reason philosophers have given for the existence of the is-ought gap is that it is just a special case of a general principle of logic: one cannot validly deduce a conclusion that applies a predicate *P*, from premises that nowhere contain that predicate.¹⁷ This is nothing special about evaluative predicates; it applies to any predicate whatsoever. For instance, one cannot infer a conclusion stating that something is red from premises that nowhere contain the word 'red'.

Before turning to objections to Hume's Law, it is important to understand what I take the doctrine's significance to be. I do not claim that Hume's Law refutes ethical reductionism.¹⁸ Rather, I see Hume's Law as an important lemma in my larger argument that one cannot account for moral knowledge without appealing to something like ethical intuition. To that end, I argue: (a) that we cannot know moral truths by observation, (b) that we cannot know (non-trivial) moral truths by deducing them from non-moral truths, (c) that we cannot know (non-trivial) moral truths by conceptual analysis, and (d) that we cannot know moral truths by scientific reasoning or inference to the best explanation. This understanding of the use I make of Hume's Law has implications for how we should treat alleged exceptions to the Law: such exceptions are important only if they represent possible ways of *coming to know* moral truths on the basis of non-moral truths. Thus, if there should prove to be a valid deductive inference containing only non-evaluative premises and an evaluative conclusion, but for some reason this inference could not be the source of our knowledge of the conclusion, then it would be irrelevant to my purposes here. It would be an exception to Hume's Law as traditionally stated, but not one that damages my argument in this chapter.

***4.3.2 Searle's challenge**

John Searle asks us to consider the following sequence of statements:

1. Jones uttered the words, 'I hereby promise to pay you, Smith, five dollars'.
2. Jones promised to pay Smith five dollars.

3. Jones placed himself under (undertook) an obligation to pay Smith five dollars.
4. Jones is under an obligation to pay Smith five dollars.
5. Jones ought to pay Smith five dollars.

(1) by itself does not entail (2); in order to conclude that Jones promised to pay Smith five dollars, we must have a body of background knowledge about the conventions in Jones' society (those that determine the meanings of Jones' words, how promises are made, etc.) and the circumstances of Jones' utterance (Jones was not joking, reading lines for a play, etc.). But this additional information consists entirely of natural facts. So let us just imagine that this information is all added to the premises of the argument, and ask whether the argument then would bridge the is-ought gap.

(3) follows from (2), Searle says, because it is part of the concept of promising that a promise is the undertaking of an obligation; this is what distinguishes promising to do something from merely predicting that one will do it or expressing an intention to do it.

From (3) it follows at least that *other things being equal* (provided nothing else happens to remove the obligation), Jones is under an obligation to pay Smith five dollars.

From (4) it follows that, other things being equal (provided there is no more important, conflicting obligation), Jones ought to pay Smith five dollars. Thus, it seems that we can at least get some such conclusion as 'Other things being equal, Jones ought to pay Smith five dollars' from non-evaluative premises.¹⁹

What is wrong with this argument? To begin with, let us distinguish two senses of 'undertook', which I will call the strong sense and the weak sense. In the strong sense, to undertake an obligation is to successfully place oneself under an obligation, such that it logically follows that one is in fact under an obligation. In the weak sense, to undertake an obligation is, roughly, to *purport* to place oneself under an obligation and/or to agree to accept such an obligation. (The exact analysis is not important here.) A person who undertakes an obligation in the weak sense need not ever actually have an obligation. Consider an analogy: suppose you call me collect, and I agree to accept the charges, but due to a mistake, the phone company never actually charges me. My having 'accepted' the charges does not entail that I ever actually receive the charges. In a similar sense, a person might conceivably 'accept' an obligation without ever actually having the obligation. Why might this happen? Well, suppose, for example, that nihilism is true: that there are no moral properties whatsoever. Then although people frequently 'accept' and 'undertake' (in the

weak sense) obligations, their undertakings would always fail to actually place them under obligations.

Of course, I believe that nihilism is false and that we do have obligations to keep our promises. But Searle has not refuted nihilism, and so he has not proven an 'ought' statement. There is no purely natural fact that the nihilist is unable to recognize. The nihilist, confronted with Searle's derivation, should grant that Jones made a promise, when this is understood as the acceptance or undertaking of an obligation in the weak sense; but he should deny that Jones undertook an obligation in the strong sense of successfully placing himself under an obligation.

Searle might argue that the weak notion of undertaking obligations does not do justice to our concept of 'promising'; if so, the nihilist should say, then no one ever really promises; at most we *purport* to promise. The conventions of our society may dictate that saying such words as 'I hereby promise to pay you five dollars' is taken as creating an obligation, but mere conventions cannot make it the case that such things as 'obligations' really exist; we cannot render nihilism false by mere fiat.²⁰

***4.3.3 Geach's challenge**

Peter Geach proposes the following alleged counter-example to Hume's Law:

1. If Evan were to promise to adopt some practice then he would adopt it. (Premise.)
2. If Evan were to utter sentence *W*, he would be promising to adopt the practice of doing something wrong twice a day. (Premise.)
3. Nobody should adopt the practice of doing something wrong twice a day. (Premise.)
4. If Evan were to utter *W*, he would adopt the practice of doing something wrong twice a day. (From 1, 2.)
5. Therefore, Evan should not utter *W*. (From 3, 4.)²¹

We imagine *W* to be some such sentence as, 'I promise henceforth to do two wrong things each day', and we imagine Evan to be a person with a very strong commitment to promise-keeping, such that (1) is true. The above appears to be a valid inference from premises (1), (2), and (3) to conclusion (5). (5) is clearly evaluative. (1) and (2) seem to be non-evaluative. If (3) were non-evaluative, the counter-example would seem to be in good order, but in fact (3) is evaluative. Nevertheless, Geach observes that (3) is an analytic truth. By definition, no one should do what is wrong; *a fortiori*, no one should do something

wrong twice a day. So at least Geach seems to have shown how to go from non-evaluative statements and/or tautologies to a non-trivial evaluative conclusion.

Geach's example has at least two flaws.²² First, the argument is invalid. I assume that the inference from (3) and (4) to (5) is valid only if the following is a valid form of inference:

6. *S* ought not to do *A*.
7. If *S* did *B*, *S* would do *A*.
8. Therefore, *S* ought not to do *B*.

To see the problem with this, imagine the following scenario. Jon is a judge about to pass sentence on Mary, a convicted marijuana dealer. Mary's crime is minor at worst; John, however, has an intense, irrational hatred for all drug users, as a result of which he is determined to sentence Mary to either life imprisonment or death. He *could* sentence Mary to only a brief prison term, but he would not in fact do so. Now consider the following inference:

9. John ought not to sentence Mary to life imprisonment.
10. If John were to refrain from sentencing Mary to death, then he would sentence Mary to life imprisonment.
11. Therefore, John ought not to refrain from sentencing Mary to death (that is, he ought to sentence Mary to death).

(9) is true because what John *ought* to do is to sentence Mary to a brief prison term at most. (10) is true because of what we have stipulated about John's psychology. But (11) is false. In general, this form of inference is invalid because it is possible for there to be a situation in which what one *should* do is to do *B* and refrain from doing *A*, but what one actually *would* do after doing *B* is to do *A* as well. In such a situation, (6) and (7) are true and (8) is false.

The second problem with Geach's example is that the premise, 'If Evan were to promise to adopt some practice then he would adopt it', is evaluative. The relevant reading of (1) is:

$(\forall x)(\text{Evan promises to do } x \rightarrow \text{Evan does } x)$

('for all *x*, if Evan were to promise to do *x*, then Evan would do *x*'). This is evaluative because the following substitution instance of it is evaluative:

Evan promises to act wrongly $\rightarrow$ Evan acts wrongly.

That is evaluative because, once we fix the natural facts about how Evan would behave upon making such a promise, whether the whole sentence is true then depends on whether that behavior would be wrong. For example, let's suppose that, if Evan were to promise to act wrongly, he would subsequently (thinking these things to be wrong) steal a piece of candy from his five-year-old niece and order a hamburger from McDonald's. Those are the natural facts about how Evan would behave. Whether 'Evan promises to act wrongly $\square \rightarrow$ Evan acts wrongly' is true then depends on whether stealing the candy or ordering the hamburger would in fact be *wrong*. If they would, then the quoted sentence is true; if they would not (and no other thing Evan would do would be wrong), then the quoted sentence is false. That is why 'Evan promises to act wrongly $\square \rightarrow$ Evan acts wrongly' is evaluative. This is an application of the criterion for evaluative statements developed below in section 4.3.5.

*4.3.4 Prior's challenge

A. N. Prior offers the following inference allegedly spanning the is-ought gap:

- (A) 1. Tea-drinking is common in England.
2. Therefore, tea-drinking is common in England, or all New Zealanders ought to be shot.²³

The inference is perfectly valid, 'ought' appears in the conclusion, but it does not appear in the premise.

We might respond by denying that the conclusion is genuinely evaluative. But this would only open up the possibility of a different inference bridging the is-ought gap:

- (A') 1'. Tea-drinking is not common in England.
2. Tea-drinking is common in England, or all New Zealanders ought to be shot.
3. Therefore, all New Zealanders ought to be shot.

If (2) is evaluative, then argument (A) spans the is-ought gap; if (2) is non-evaluative, then (A') spans the is-ought gap.

However, this point has little bearing on the problem of moral knowledge, as neither of these inferences provides a plausible general model of moral knowledge. Although (A) may give us knowledge of an 'evaluative conclusion' in some sense, it is not an interesting evaluative conclusion. It is not one that, for example, we might use to guide our action, nor does it entail that anything actually has any

evaluative property. The conclusion of (A'), on the other hand, is an interesting evaluation—but (A') provides no model for moral knowledge since there is no obvious way in which we could know both of its premises. If (1') were true, then we could readily discover it by empirical means, but there is no apparent way in which we could simultaneously discover (2) to be true without relying on an evaluative assumption.

What would be helpful would be an argument that combined the virtues of (A) and (A'): an argument that started from something obviously empirically verifiable, like (1), and concluded with an interesting, action-guiding evaluation, like (3). But Prior's example suggests no way of constructing such an argument. Obviously, we cannot simply combine the premises of (A) and (A'): that would indeed give us a valid argument from non-evaluative premises to an evaluative conclusion. But since the premises would be inconsistent, this argument would provide no plausible model of how we come to know moral truths.

***4.3.5 Karmo's proof**

The preceding response to Prior may leave one worried: once it has been shown that the is-ought barrier is not in principle impregnable, how can we be sure that every argument penetrating that barrier will prove as useless in accounting for moral knowledge as Prior's example? Can we devise a general proof for some useful form of Hume's Law?

Toomas Karmo has provided such an argument.²⁴ Assume that there is both a class of *uncontroversially* evaluative statements which clearly attribute evaluative properties to things—such as 'All New Zealanders ought to be shot' and 'Pleasure is good'—and a class of uncontroversially descriptive statements which clearly attribute only non-evaluative properties—such as 'Tea-drinking is common in England'. There are also some statements whose status is initially unclear, such as 'Tea-drinking is common in England, or all New Zealanders ought to be shot' and 'Everything Alfie says is true'. The latter statements, Karmo says, may be evaluative or non-evaluative, depending on what the world is like in non-evaluative respects. For instance, if tea-drinking is in fact common in England, then 'Tea-drinking is common in England, or all New Zealanders ought to be shot' is *non-evaluative* (roughly, because its truth does not then depend on whether New Zealanders ought to be shot); otherwise, the statement is evaluative. Similarly, if Alfie has in fact made some evaluative statements and has made no false descriptive statements, then 'Everything Alfie says is true' is evaluative, since its truth then

depends on whether Alfie's evaluative statements were true; otherwise, it is non-evaluative.

Following is a more precise statement of Karmo's idea. Let a 'value system' be a maximal consistent conjunction of uncontroversially evaluative statements (that is, a conjunction of uncontroversially evaluative statements that is consistent and such that no further such statements could be added to it without creating an inconsistency). Similarly, let a 'possible world' be a maximal consistent conjunction of uncontroversially descriptive statements. The 'actual world' is the possible world that is true (the correct description of the world). I assume the following principle, which I shall call the Determinacy Assumption:

Determinacy A possible world together with a value system are consistent with at most one assignment of truth-values for all statements.

That is, if w is a possible world, v is a value system, and S is any statement, then either $(w \ \& \ v)$ entails S , or $(w \ \& \ v)$ entails $\sim S$.²⁵

Now, for statements whose status as evaluative or descriptive is initially unclear, classify them as follows: a statement S is evaluative in possible world w if, given w , the truth-value of S depends on which value system is correct. In other words, S is evaluative in w provided that there are two value systems such that one of them, combined with the uncontroversially descriptive statements in w , implies that S is true, while the other, combined with the same descriptive statements, implies that S is false. This characterization yields the result, for example, that 'Everything Alfie says is true' is evaluative in any possible world in which Alfie has made an evaluative statement and has made no false descriptive statements, but non-evaluative in any other possible world.

What we would like to see is whether there can be an argument such that

- i) Its premises are actually true,
- ii) Its premises are non-evaluative in the actual world,
- iii) Its premises entail its conclusion,
- iv) Its conclusion is evaluative in the actual world, and
- v) Its conclusion is non-trivial.

Conditions (ii), (iii), and (iv) are required for the argument to count as spanning the is-ought gap. Condition (i) is required for the argument to count as a way of gaining knowledge. Condition (v) is

required for the argument to be interesting vis-a-vis the problem of moral knowledge—the conclusion should not be a mere tautology such as ‘Right acts ought to be performed’ or ‘Bad things are bad’ (these statements trivially follow from any set of premises). I assume that an evaluative statement is non-trivial only if there is at least one value system that rejects it.

Now assume for *reductio* that there is an argument satisfying conditions (i)-(v). Let $P_1, P_2, \dots$ be the premises of this argument, and let C be its conclusion. Let w be the actual world. Then we have the following:

1. $P_1, P_2, \dots$ entail C . (Condition (iii).)
2. C is non-trivial and evaluative in w . (Conditions (iv) and (v).)
3. There exist value systems v_1 and v_2 such that $(w \ \& \ v_1)$ entails C and $(w \ \& \ v_2)$ entails $\sim C$. (From 2.)
4. $\sim C$ entails that at least one of $\{P_1, P_2, \dots\}$ is false. (From 1.)
5. $(w \ \& \ v_2)$ entails that at least one of $\{P_1, P_2, \dots\}$ is false. (From 3, 4.)
6. There is a specific premise in $\{P_1, P_2, \dots\}$, call it P_i , such that $(w \ \& \ v_2)$ entails $\sim P_i$. (From 5 and Determinacy.)
7. w conjoined with the correct value system entails P_i . (From condition (i) and Determinacy.)
8. P_i is evaluative in w . (From 6, 7.)

Conclusion (8) follows from (6) and (7) because, given w , the truth of P_i turns on what the correct value system is— P_i is true according to the correct value system but would be false according to v_2 . But (8) contradicts condition (ii). Thus, there is no argument satisfying all of conditions (i)-(v).

I take it that Karmo’s proof succeeds. However, Stephen Maitzen has objected to the argument.²⁶ Without going into detail, I note that Maitzen’s objection turns on a criticism of Karmo’s taxonomy, a criticism that depends on the assumption that nihilism—defined as the thesis that every evaluative statement is false—is logically possible. Among other things, Maitzen argues that Karmo’s taxonomy is defective since it implies that the following statement is evaluative in every possible world:

B. Every evaluative statement is false, or torturing babies is wrong.

According to Karmo’s taxonomy, this is evaluative in every possible world, since there is always a value system that makes the statement true (any value system that forbids baby-torture makes (B) true) and another value system that makes the statement false (any value

system that endorses baby-torture will make both disjuncts of (B) false). But Maitzen says the statement should be classified as non-evaluative in at least some possible worlds, for:

1. Nihilism is logically possible. (Premise.)
2. Nihilism entails (B). (By $\vee$ -introduction.)
3. Therefore, it is possible that nihilism and (B) both hold. (From 1, 2.)
4. In any possible world in which nihilism and (B) both hold, (B) is non-evaluative. (In such a world, every evaluative statement is false, since nihilism holds. But (B) also holds, so (B) must not be an evaluative statement.)
5. So there is at least one possible world in which (B) is non-evaluative. (From 3, 4.)

This example, Maitzen says, ‘immediately falsifies Karmo’s taxonomy of ethical sentences’.²⁷

Maitzen is mistaken because nihilism—understood as the thesis that every ‘evaluative statement’ *in Karmo’s sense* is false—is logically impossible. One reason for this is that nihilism itself seems to be an evaluative statement;²⁸ therefore, in denying all evaluative statements, it contradicts itself. Another reason is that in any possible world, both ‘Torturing babies is wrong’ and ‘It is not the case that torturing babies is wrong’ are evaluative statements in Karmo’s sense. But it is logically necessary that at least one of these two statements be true. So it is logically impossible that all evaluative statements be false. Of course, nihilism in some *other* sense might be logically possible—for example, perhaps it is possible that nothing has an evaluative property. But this would not refute Karmo’s argument. Similarly, it might be that Karmo’s sense of ‘evaluative statement’ differs from established usage—certainly it differs from Maitzen’s usage—but this would not show Karmo’s argument to be unsound either. In general, there is no question of ‘falsifying’ Karmo’s characterization of evaluative statements, as long as that characterization is understood—as I believe it should be—as a stipulative definition.

Given that the definition is purely stipulative, one may legitimately wonder whether Karmo’s proof is relevant to what we were initially interested in—in particular, does Karmo’s version of the is-ought gap help to establish that there is a problem in accounting for moral knowledge? I think it does. Karmo shows that there is no way of knowing evaluative statements by deduction from non-evaluative statements, in his sense of ‘evaluative statements’. But Karmo’s sense

includes all the statements traditionally considered 'evaluative' by moral philosophers; thus, if it is difficult to know evaluative statements in Karmo's sense, it is also difficult to know evaluative statements in the traditional sense. Nor does Karmo appear to have expanded the sense of 'evaluative statement' so much that we might know some evaluative statements, for example, by observation: in Karmo's sense, an evaluative statement is one whose truth, once all the descriptive facts are fixed, still turns on which value system is correct. Barring appeals to such things as moral intuition, it remains difficult on the face of it to see how we might know such statements non-inferentially. Since Karmo has shown that we also cannot in general know such statements by deduction from non-evaluative statements, there is an interesting problem of moral knowledge.

In the next section, we consider attempts to show that we can know evaluative truths by *non*-deductive inferences from non-evaluative truths.

4.4 Synthetic reductionism

Synthetic Reductionists appear to escape the Open Question Argument.²⁹ They maintain that there is some true statement of the form, 'For a thing to be good is . . . ', where the ellipsis is filled in using non-evaluative terms. But they admit that as Moore showed, what fills in the ellipsis will not be *synonymous* with 'for it to be good'. Consider some analogous cases:

For a thing to be hot is for its molecules to have high average kinetic energy.

For a substance to be water is for it to consist of H₂O molecules.

For a thing to make a sound is for it to cause a compression wave in the surrounding medium.

Quibbles aside, all of those statements are true. But they cannot be verified solely by considering the meanings of the words 'hot', 'water', and 'sound' in English. People knew these words and used them in the same sense we presently do for centuries without knowing the facts expressed by the above statements. Very specialized observations and scientific reasoning were required to confirm those facts. Just so, says the synthetic reductionist, we will someday find (if we have not already found) a theory of what it is for something to be good, and that theory will not be known simply on the basis of the meaning of the word 'good' in English. Instead, it will be

known in a way analogous to how the above statements are known to be true, by a kind of inference to the best explanation.³⁰

In this section, I argue that the synthetic reductionists' account of moral knowledge fails: no theory about the nature of goodness can be known in any manner analogous to how the above theories of heat, water, and sound are known. In the end, the synthetic reductionist will have to appeal to ethical intuition, just as the intuitionists do. This deprives their position of one of its central alleged advantages.

4.4.1 Can moral facts be known by observation?

The analogy just drawn between ethical reductionism and various scientific theories plays a fairly serious role in motivating synthetic reductionism. But the analogy fails at a crucial point.

Suppose we are speaking to a synthetic reductionist who holds that for a thing to be good is for it to be N , where N is some natural property, or for short:

Good = N .

The reductionist says this is analogous to the theories mentioned above, which I shall abbreviate as follows:

Heat = Molecular kinetic energy.

Water = H_2O .

Sound = Compression waves.

In each of these cases, we have two *different concepts* that refer to the *same phenomenon* (the concept 'water' is different from the concept ' H_2O ', but there is one thing in reality that both these concepts pick out). In each case, the concept on the left hand side is observational (we can apply it to a thing on the basis of observation), while the concept on the right hand side is theoretical (we apply it to a thing on the basis of inference, including an inference to a scientific theory) and is taken to *explain the nature of* the phenomenon in question. Thus, we knew that things were *hot*, initially, because we could feel heat with our hands. We know that things have *molecular kinetic energy* on the basis of complex inferences from observational data to the molecular theory of matter, and so on. The idea of molecular kinetic energy is thought to explain the underlying nature of heat. Similarly, we know that there is water on the basis of casual observation, but we know that there is H_2O on the basis of complex

scientific reasoning. And 'H₂O' explains the underlying nature of water.

Now, in the reductionist's analogy, which term is 'good' supposed to be analogous to: the term on the left hand side of the '=', or the term on the right? Is 'good' supposed to be like 'heat' or like 'molecular kinetic energy'? Initially, it seems that 'good' is supposed to be analogous to 'heat', because the reductionist's theory is supposed to explain the underlying nature of goodness in terms of natural properties, just as the scientific theory explains the underlying nature of heat in terms of motions of molecules. Certainly ethical reductionists do not take themselves to be explaining the underlying nature of some natural properties in terms of moral properties.

This suggests that moral concepts should be observational, just as 'heat', 'water', and 'sound' are. The reason why we believe, for example, that water = H₂O is, roughly, that (i) we have independent (that is, pre-theoretical), direct awareness of the presence of water, and (ii) the theory that this substance of which we are aware is composed of H₂O molecules helps explain many of its observable features (again, features of which we know independently of our scientific theory). If we are to take the synthetic reductionist's analogies seriously, then, we should say that we have independent, direct awareness of the presence of goodness, and that the theory that goodness is identical with *N* helps explain many of the observable features of goodness.

The problem is that in fact, moral properties are entirely unobservable.³¹ Moral value does not look like anything, sound like anything, feel (to the touch) like anything, smell like anything, or taste like anything. The only plausible way to maintain that we have direct awareness of moral facts would be to appeal to either 'ethical intuition' or a 'moral sense'—which is precisely the sort of 'mysterious', 'spooky' thing the reductionist wants to avoid.

Gilbert Harman discusses an alleged example of moral observation.³² You see a group of hoodlums pour gasoline on a cat and ignite it. You immediately think that what they are doing is wrong. According to Harman, 'you do not need to figure anything out; you can see that it is wrong'. But this quotation is misleading by itself; Harman does not think that the wrongness of the action somehow modifies the light rays traveling to your eyes, nor does he think that your alleged ability to see wrongness ultimately explains your moral knowledge. Rather, he thinks that you have background moral beliefs, or a moral sensibility, which cause you, upon seeing an event such as the one described, to immediately (without thinking about it) form the belief, 'That is wrong'. Since the 'seeing' of wrongness

depends upon prior moral beliefs, it does not aid in the ultimate explanation of how we know moral truths.

Why does Harman consider this a case of observation at all? He seems to think that to 'observe' that *p* is just to form a belief that *p* as a fairly direct result of perception.³³ This is mistaken; a belief formed as a fairly direct result of perception need not be an *observation*. Suppose I am superstitious and whenever I see a black cat, I immediately (without thinking about it) believe that an airplane is going to crash the next day. I am now seeing a black cat. Do I thereby observe a plane crash? No. More is required for observation. For someone to observe that an object is *F*, where *F* is some property, there must be *a way that F things look* (or sound, smell, etc.), and the object must look (sound, smell, etc.) that way.³⁴ For instance, I have a pair of artificial leather shoes. Most people can perceive by touch that they are not leather, because there is a way that real leather feels, and these shoes do not feel that way. The point of interest here is that there is no such thing as *the way that wrongful actions look* or *the way that permissible actions look*. That is why you cannot literally see, with your eyes, that an action is wrong.³⁵

Consider another argument for the possibility of moral observation. Michael Moore argues that since moral properties *are* natural properties (per his ethical naturalist stance), and since we may be able to perceive the relevant natural properties, we can therefore perceive moral properties. In short, Moore takes the following argument to be valid:

I can perceive *N*.

Good = *N*.

Therefore, I can perceive goodness.³⁶

Before addressing this, I must make a small correction: the relevant question is not whether one can *perceive goodness*; it is whether one can *perceive that a particular thing is good*. These are two different questions, and only the latter bears on the problem of moral knowledge. To illustrate, compare this case: I can see your car (your car reflects light to my eyes, and so on) without being in a position to know that the car is yours. If I could see *that it is your car*, that would enable me to know that it is yours, but merely seeing your car doesn't give me that knowledge. So let us modify Moore's argument as follows:

I can perceive that *x* is *N*.

Good = *N*.

Therefore, I can perceive that *x* is good.

If this inference were valid, Moore would have a viable account of how we can know moral truths. But in fact, the inference is invalid. It has the same form as the following inference:

I can perceive that x is hot.

Heat = Molecular kinetic energy.

Therefore, I can perceive that x 's molecules have high average kinetic energy.

Here, I use one of the reductionists' examples against them. The first two premises may be true, but the conclusion is certainly false. For instance, one can perceive by touch that a stove is hot. But one does not, merely by feeling it, perceive that the stove is composed of fast-moving molecules (even though that is really what it is for the stove to be hot). One has to do some sophisticated theorizing to find that out. Similarly, even if $\text{Good} = N$, and even if one can perceive that x is N , one does not therefore perceive that x is good.³⁷

Value facts, then, are unobservable. This is a problem for the synthetic reductionist's analogies: we were able to discover that heat = molecular kinetic energy, only because we could first identify which things were hot by observation. If we have no pre-theoretical knowledge of which things are good, then we cannot discover that $\text{Good} = N$ in any analogous way. The situation would be like the following scenario: someone tells you that there is an unobservable property called 'torfness'. You are given no initial information about the nature of torfness or which things in the world, if any, are torf. You are told only that torfness is reducible to some scientific property. Your assignment: figure out which scientific property torfness is reducible to.

It seems clear that you do not have enough information to carry out the assignment. Likewise, with no initial information about the nature of goodness or which things have it, we would have no basis for framing any hypotheses as to what natural property goodness might be reducible to. As we have just seen, observation furnishes us with no such initial information. Nor, according to the naturalists, does intuition. One might be tempted to propose that we simply rely on our common sense beliefs about good and bad. But unless we are justified in thinking these beliefs are generally true, this would be arbitrary, and we have as yet seen no account of why those beliefs are justified. Moreover, if the naturalist could give an account of our initial knowledge of the nature of value or of which things are good, he wouldn't need the story about inference to the best explanation that naturalists have been at such pains to develop (see subsections 4.4.2–4.4.4).

4.4.2 Can moral facts be known by inference to the best explanation?

Some reductionists have argued that we are justified in believing moral claims because they provide the best explanations for certain non-moral facts of which we have independent awareness. For instance, the best explanation for why Hitler ordered the Final Solution is that Hitler was depraved; only a depraved person would do such a thing.³⁸ Furthermore, since we know empirically that Hitler ordered the Final Solution, and our justification for believing he was depraved is that that explains why he ordered the Final Solution, our knowledge that he was depraved is also empirical. In short:

1. Moral claims help explain observable facts. For instance, Hitler's depravity helps explain why he ordered the Final Solution.
2. If a claim helps to explain observable facts, then we have empirical justification for believing that claim.
3. Therefore, we have empirical justification for believing some moral claims, such as that Hitler was depraved.

The conclusion in (3) is to be understood as implying that we can have *purely* empirical justification for believing some moral claims—that is, justification that does not ultimately depend, even partly, upon intuition or any other non-observational source of information.

Most of the discussion in the literature centers on premise (1). There are two main arguments for (1). I call them 'the causal argument' and 'the counterfactual argument'.

The causal argument holds that for x to explain y is for x to be (part of) the cause of y . Hitler's ordering of the Final Solution was caused by certain psychological traits of his, such as extreme racism, hatred, and a lack of respect for human life. According to the ethical reductionists, these traits *constituted* Hitler's depravity. Therefore, Hitler's depravity caused his ordering of the Final Solution. (This argument employs the valid principle that if A causes C , and B is identical with A , then B causes C .) Therefore, Hitler's depravity explained his order.³⁹

The counterfactual argument holds that x explains y (at least in part) provided the following is true: if x had not occurred, y would not have occurred. Now, if Hitler had not been depraved, he would not have ordered the Final Solution. Since moral facts are reducible to natural facts, if Hitler had not been depraved, he would have to have had different psychological traits from those he actually had—for instance, he would have to have not hated the Jews, or have had

more respect for human life, etc. And if that were the case, then he would not have ordered the Final Solution. So his depravity explains (at least in part) why he ordered the Final Solution.⁴⁰

I agree that moral facts can be explanatory; premise (1) is true. But premise (2) is false. To see why, return to the reductionists' favorite analogy: the theory that water = H_2O . Imagine a scientist attempting to justify that theory by the means suggested by ethical reductionists. Suppose Lavoisier had published a paper arguing that we are justified in believing that water is composed of hydrogen and oxygen, on the grounds that the presence of such a compound explains certain observable facts. For instance, I have an observable stain on my coffee table that was caused by water. Since water = H_2O , we can also say that the stain was caused by H_2O . Therefore, H_2O explains an observable fact. Therefore, we have empirical justification for believing in H_2O .

Here is another argument. If H_2O wasn't spilled on my coffee table, there wouldn't be a water stain there now. For remember, water consists of H_2O , so in order for H_2O not to have been on my coffee table, there would have to have been no water on my coffee table. And if that were the case, there would not be a water stain there now. Thus, the (past) presence of H_2O explains a currently observable fact. So again, we are justified in believing in H_2O .

Thankfully, this is not how scientists reason. Here is part of why scientists believe water = H_2O : if you apply an electric potential to a sample of water, bubbles of hydrogen form at the cathode. Bubbles of oxygen form at the anode. The mass of water present decreases by the same amount that the mass of hydrogen and oxygen present increases.⁴¹ Furthermore, two parts hydrogen are formed for every one part oxygen (by moles). That is just one of several relevant pieces of experimental evidence, but that will do for illustrative purposes. The moral theorist has no empirical evidence to offer that is analogous to that.

This example suggests that the causal and counterfactual arguments do *not* show how moral claims can be justified, since analogous arguments would not work for scientific claims. But exactly what is wrong with those arguments?

Begin with the causal argument. If ethical reductionism is true, then *moral properties* can cause observable effects. Nevertheless, this would not explain how we know moral claims. For us to know a moral claim on the basis of inference to the best explanation, the *moral claim* would have to explain some observable fact *that could not otherwise be explained*. For instance, we can know that Hitler was depraved by inference to the best explanation, if the claim that Hitler

was depraved provides an explanation for some fact that cannot be explained without invoking moral claims. But in fact, the hypothesis of Hitler's depravity does not explain any new facts about his behavior that aren't already explained by non-moral claims about him, such as that he hated Jews, that he had a lust for power, that he lacked respect for human life, and so on. Since the truth of non-moral claims of that kind is not in dispute, the moral explanation is superfluous and provides no justification for adopting any moral theories. This is not to say, of course, that we have no justification for our moral claims about Hitler; if we were *already* justified (perhaps on the basis of moral intuition) in accepting moral theories such as that mass murderers are evil, then we can infer that Hitler was evil. The point is that neither our general moral theories nor our moral judgments about Hitler can be established solely on the basis of observation.

The same applies to the counterfactual argument. Given that racial hatred and so on constitute depravity, it is true that had Hitler not been depraved, he would not have acted as he did. But we can explain Hitler's behavior equally well without saying that racial hatred and so on constitute depravity: had Hitler lacked racial hatred, power lust, and so on—whether or not these things constitute depravity—he would not have acted as he did. So moral assumptions add no explanatory power.

To confirm this diagnosis, recall the water/H₂O example. The water stain on my coffee table fails to confirm the past presence of H₂O, since the hypothesis that H₂O was spilled on the table does not explain any *new* fact that was not already explicable merely by the hypothesis that water was spilled on the table. But the electrolysis experiment confirms that H₂O is present, because the hypothesis that H₂O is present explains where the hydrogen and oxygen are coming from. That is *not* explained merely by the hypothesis that *water* is present, without assuming that water is composed of hydrogen and oxygen.

***4.4.3 Can moral claims be tested?**

Perhaps moral claims are justifiable because they make empirical predictions that can be tested. For instance, the two moral claims 'Hitler was an admirable person' and 'No admirable person would kill millions of people' together entail the following prediction: 'Hitler would not kill millions of people'. This prediction is testable and, as a matter of fact, false. So we see that moral claims are testable just as scientific theories are.

In case you are tempted to object that neither moral claim *by itself*

entails any testable predictions, Nicholas Sturgeon reminds us that it is also true that individual scientific claims do not, in general, entail any testable predictions by themselves; normally a scientific theory makes a testable prediction only given a large background of other scientific knowledge.⁴²

Even granting the last point, it does not seem that this accounts for moral knowledge. Imagine that Lavoisier had argued analogously: Collections of claims about H_2O entail testable predictions. For instance, the pair of claims 'Water = H_2O ' and ' H_2O boils at 212°F' together entail the testable prediction, 'Water boils at 212°F'. Since the latter prediction is in fact true, my H_2O beliefs thereby receive confirmation.

The problem is that not all predictions made by a theory create ways the theory can be usefully confirmed. If each prediction is made with the help of a separate *ad hoc* posit, then the theory's success at prediction will be unimpressive. Thus, the 'Water = H_2O ' theory can make a series of testable predictions about the nature of water, if we add to it the *ad hoc* suppositions that H_2O would be a clear, odorless liquid, that it would boil at 212°F, and so on; but the success of these predictions is evidentially nugatory. In contrast, the idea that water = H_2O naturally suggests that water should be decomposable into hydrogen and oxygen in a 2:1 ratio; this prediction requires no *ad hoc* posits.

The predictions made by moral theories all appear to require *ad hoc* posits. The moral theory that killing millions of people is wrong enables us to predict that Hitler won't kill millions of people, only if we make the unmotivated assumption that Hitler is morally admirable (or at least not evil). But the possibility of deriving predictions from an *arbitrary* combination of moral statements is of no epistemological interest.

Sturgeon gives another example. The pair of claims, 'Injustice causes political instability' and 'Natural property X constitutes injustice' together entail that X causes instability.⁴³ Provided X is independently detectable, this is a testable prediction. Here, Sturgeon's assumption that injustice causes instability does not seem *ad hoc*; it seems like a natural assumption about the nature of injustice. But why? Presumably because we assume (a) that people can typically identify injustice, and (b) that they object to it. Given these assumptions, when injustice is widespread in a society, it is natural to expect that the society will be politically unstable. If people had no means of moral knowledge to begin with, then there would be no reason to expect that moral properties would be correlated to their behavior in any interesting way.

The problem now is that our ability to make a prediction based on moral claims in this case depends on our assuming that people already have some way of gaining reliable moral beliefs. And what might that be? If we accepted that people have a reliable faculty of moral intuition, say, or that people can observe moral facts, or that they can deduce moral facts from non-moral facts, then we could accept this. But the synthetic reductionist rejects the first possibility, and we have rejected the latter two already. Moreover, if the reductionist could answer this question, then he wouldn't *need* the present argument—he would already have an account of moral knowledge.

In sum, if we are given all the non-moral properties of a thing, the additional hypothesis that one of those properties constitutes *goodness*, say, does not seem to naturally generate any *new* predictions about what we should observe, without a separate *ad hoc* posit for each desired prediction. Consequently, starting from a system of purely non-moral beliefs, we would not be justified in adding any moral beliefs. There is one exception: it seems natural and not *ad hoc* to expect that intelligent beings will by and large pursue the good, feel gratified when it is obtained, and so on. But this prediction depends upon the prior assumption that people have some independent means of detecting moral properties, so it is of no use in accounting for the origin of moral knowledge.

***4.4.4 The unifying power of moral explanations**

Perhaps, despite the fact that they only provide explanations for things we already have non-moral explanations for, moral claims can still be justified because they provide *better* explanations than those formulated using non-moral claims. Michael S. Moore has developed this suggestion, arguing that moral explanations can achieve a kind of unification not otherwise available.

An analogy will help. We find that a certain peg will not fit into a certain hole. On one level, we could try to explain this fact by citing the trajectories of all the atoms in the block and the chemical and other forces acting on each of them. But a more perspicuous explanation would be: 'This is a square peg, and that's a round hole'. The latter explanation is superior in its unifying power: given this explanation, we know what to expect when we try putting another square peg into another round hole (with a diameter less than the diagonal of the square). The atomic-level explanation misses this higher-level regularity: the next square peg we encounter may be made of a different substance and will certainly have a different arrangement of atoms, so on the atomic level, the same explanation

will not apply to it; we will have to give a distinct explanation in each case. In contrast, the 'square peg, round hole' explanation identifies the essential feature common to all these situations.⁴⁴

Now, suppose the property of goodness is identical with some disjunction of natural properties, that is,

$$\text{Good} = (F \text{ or } G \text{ or } H),$$

where '*F*', '*G*', and '*H*' are non-evaluative terms. And suppose there is some effect, *E*, which occurs whenever *any* of *F*, *G*, or *H* is present. In that case, although the occurrence of *E* on a given occasion might be explained by citing the presence of *F*, the *better* explanation would be the one that cites the presence of goodness. For the latter explanation would achieve a kind of unification the former explanation does not—it enables one to correctly predict what will happen when another instance of goodness occurs.⁴⁵

The crucial question here is, how would one come up with the idea that $\text{Good} = (F \text{ or } G \text{ or } H)$? Presumably, the mere fact that each of several distinct properties brings about a single effect does not justify identifying goodness with the disjunction of those properties. One might be tempted to appeal to common sense beliefs about which things are good—but that would be to revert to a mistake we have earlier criticized. Since at this point the reductionist has not yet successfully explained how people can access moral reality, he is not entitled to assume that our common sense moral beliefs are reliable. Since no advantage of the proposed moral explanation has been identified other than its unifying power, any equally simple explanation that achieves the same unifying power ought to be equally acceptable. Thus, it would apparently make just as much sense to propose that $\text{Bad} = (F \text{ or } G \text{ or } H)$. Or why not invent some other, non-evaluative concept, say 'torfness', and say that $\text{Torf} = (F \text{ or } G \text{ or } H)$ —why not, in other words, say that $(F \text{ or } G \text{ or } H)$ is identical with a single property, but that property is not evaluative?

Synthetic reductionists concede that the *meaning* of an evaluative term is not captured by any non-evaluative expression. What the questions of the last paragraph illustrate is that the distinctive, positively evaluative meaning of 'good' plays no role in the sort of explanations the reductionist would offer us—since all that matters is that *F*, *G*, and *H* be brought under a single concept, a negatively evaluative concept or a non-evaluative concept would serve just as well. This stands in contrast with legitimate scientific explanations. When the hypothesis that $\text{water} = \text{H}_2\text{O}$ explains the result of the electrolysis of water, the specific meaning of ' H_2O ' ('a compound of

two hydrogen atoms and one oxygen atom') plays an essential role. One cannot substitute some entirely different concept, such as 'NaCl', in the theory and still produce an equally good explanation—the hypothesis 'water = NaCl' does not explain the experimental result.

4.5 The argument from radical dissimilarity

One argument against ethical reductionism that has received little attention is the argument that, from our grasp of evaluative concepts, we can simply see the falsity of reductionist theories. On the face of it, for example, *wrongness* seems to be a completely different *kind* of property from, say, *weighing 5 pounds*. In brief:

1. Value properties are radically different from natural properties.⁴⁶
2. If two things are radically different, then one is not reducible to the other.
3. So value properties are not reducible to natural properties.

If this argument sounds thin to you, it is probably because you accept one of the following two responses.

First, a reductionist might say that we cannot simply *see* the nature of a thing merely because we have a concept of it. We could not see, merely from our concept of water, that water was identical with H₂O. But my argument doesn't assume that we can see *everything* about the nature of those things we have concepts of; it only assumes that we can see *some* things about them, and that in the case of physical and psychological properties, we can see that they are not evaluative. To illustrate, suppose a philosopher proposes that the planet Neptune is Beethoven's Ninth Symphony. I think we can see that that is false, simply by virtue of our concept of Neptune and our concept of symphonies. Neptune is an entirely different kind of thing from Beethoven's Ninth Symphony. No further argument is needed. Indeed, if a person *couldn't* see that Neptune is not a symphony, we would say he either had no idea what Neptune was, or had no idea what a symphony was.

Second, a reductionist might say that the argument, if accepted, would also 'refute' the theory that Water = H₂O, for 'water' and 'H₂O' just seem very different on their face. Similarly, the argument would seem to refute the theory that sound consists of compression waves in the air, because 'sound' and 'compression waves' seem very different.

But this is not so. On the face of it, water *doesn't seem to be* H₂O; but it is not the case that water *seems to not be* H₂O. Our pre-scientific

concept of water takes it to be a clear, odorless, tasteless, etc., liquid. We cannot, on the basis of this concept, discern anything about what its micro-structure might be like. We have no experience of just seeing that there are no tiny particles composing it, and so on. Nor do we have the sense that water is a different *category* of thing from H₂O; on the contrary, we see that they are the same category of thing, namely, physical substances.

I would likewise deny that we have an experience of seeing that sound doesn't consist of compression waves, or that they are different categories of things. We can see that they are the same category of thing: physical events. I suggest that *if* you have the intuition that sound obviously can't be a compression wave in the air, then it is because by 'sound' you actually mean *the experience of sound*, which really is a different kind of thing from the compression waves in the air.

The point about the broad category that a thing belongs to is important, because in my view, our having, at least roughly and for the most part, correct intuitions or beliefs about the nature of *x* is a precondition on our referring to *x*. I can talk about Plotinus without knowing much about him; about all I know is that he was a philosopher who lived a long time ago. But I could not talk about Plotinus if I had completely inaccurate beliefs (allegedly) about him; for example, if I thought 'Plotinus' referred to a cake, then I would not be talking about Plotinus (the person) at all.⁴⁷ Similarly, if 'good' seems to us to refer to something of a fundamentally different category from natural properties, then when we say 'good' we are not talking about a natural property.

4.6 Explaining moral beliefs

I claim to have shown, in sections 4.2 and 4.5, that evaluative properties cannot be reduced to natural properties. In sections 4.3 and 4.4, I have shown that moral knowledge is not empirical. The structure of this latter argument was as follows:

1. Moral claims are known empirically only if they are known by observation or by either deductive or scientific reasoning from observations. (Premise.)
2. Moral claims are not known by observation. (§4.4.1)
3. Moral claims are not known by deduction from non-moral premises. (§4.3)
4. Moral claims are not known by scientific reasoning from non-moral premises. (§4.4)

5. Therefore, moral claims are not known empirically. (From 1, 2, 3, 4.)

I want to make one final, general point about the prospects for reductionism in ethics. The reason why we believe that such phenomena as heat, water, and sound are reducible is that we have compelling scientific arguments for identifying these phenomena with *specific* theoretical entities. No one doubts the existence of the relevant theoretical entities, because of the systematic power of the scientific theories they belong to in explaining and predicting a host of observable phenomena. Without those theories, no one would believe reductionism about heat, water, or sound.

In ethics, we have no comparable theories in hand. Reductionists typically only *predict* that moral claims will prove to be reducible.⁴⁸ They theorize in the abstract about how it might be possible to justify moral beliefs, but without presenting a detailed argument for any specific moral belief. In fact, moral philosophers have labored for two millennia without achieving a consensus on any general moral theory. In such a situation, the reductionist's position raises two questions. The first and more obvious is why we should believe that a reduction of moral terms is possible.

The second is how the reductionist explains people's actual moral beliefs. Even if one of the proposals discussed in section 4.4 were viable, none of them corresponds to how we actually form moral beliefs. If 'good' is a theoretical term like 'H₂O', shouldn't we expect that the concept would be developed only concurrently with the theory of which it is a part—and that identifiable arguments would have been given before that theory was accepted? Any chemist can identify uncontroversial examples of the sort of observations that molecular theory is needed to explain. Why are so many people unable to identify any of the observations that their moral beliefs are needed to explain? Many professionals in the field deny that there *are* any such observations, and even ethical reductionists will likely admit that the empirical reasoning behind moral theories is subtle and has yet to be clearly articulated. Yet moral beliefs abound in every society in the world, among the young and the old, the educated and the uneducated. Why? Rather than declaring that nearly all moral beliefs are as yet unjustified, shouldn't we look for some much simpler means of gaining moral knowledge?